

LPR381 Programming Project - Proposed 4-Person Work Split

Standalone meeting handout

1. Recommended Development Setup

Language: C# with .NET in Visual Studio. **Application:** either a Console App or Windows Forms App. The team should choose one before development starts. **Version control:** one shared GitHub repository with feature branches and pull requests.

2. Shared Rules Everyone Should Follow

- The input text-file format is already specified in the project brief. We only need to build a parser that reads that format correctly, not design a new format.
- The parser must support max/min, any number of variables and constraints, relations ($=$, $\leq$, $\geq$), RHS values, and sign restrictions (+, -, un, int, bin).
- Use shared model, constraint, tableau and solution classes so that every algorithm works with the same data structures.
- Keep full precision during calculations and round displayed/exported decimal values to 3 decimal places.
- All sections must combine into one working Visual Studio project, one solve.exe, and the required output text file.

3. Work Split if Everyone Presents Their Own Section

Use this split if the video is divided between all four members. Because presentation time is shared, the coding workload can also be shared more evenly.

Person	Main Area	Responsibilities
1	Core App + Simplex	Set up the Visual Studio solution and shared classes; implement the supplied-format input parser and output writer; Primal Simplex; Revised Primal Simplex; menu/interface; assist with final integration.
2	Branch & Bound Simplex	Implement LP relaxation branching, sub-problems, backtracking, fathoming, node tracking, all required tableau iterations, best candidate, and algorithm-specific validation/error handling.
3	Integer Algorithms	Implement Cutting Plane and Branch & Bound Knapsack, including cut generation, repeated solving, branching, backtracking, fathoming, node output and best-candidate reporting.
4	Sensitivity + Duality	Implement all required sensitivity-analysis operations, shadow prices, Duality, solving the dual, and strong/weak duality verification.

4. GitHub and Integration Plan

- Create one private GitHub repository and add all four members as collaborators.

- Use separate feature branches for each member or module, then merge through pull requests.
- Commit small, working changes with clear messages and pull the latest main branch before merging.
- Agree on the shared classes and method inputs/outputs before everyone starts coding separately.
- Integrate regularly. Do not leave the first full merge until the final week.

5. Video Submission Plan

The video must show the working program meeting the criteria, not only explain the code. The team should decide between these two approaches at the meeting:

- Everyone presents their own section

- Person 1: introduction, input file, interface, Primal Simplex, Revised Primal Simplex and output file.
- Person 2: Branch & Bound Simplex.
- Person 3: Cutting Plane and Branch & Bound Knapsack.
- Person 4: Sensitivity Analysis and Duality.
- One volunteer combines the clips, checks audio/order, removes unnecessary pauses, and prepares the final submission.

Suggested video order: load input file -> show model/canonical form -> Primal Simplex -> Revised Simplex -> Branch & Bound Simplex -> Cutting Plane -> Branch & Bound Knapsack -> Sensitivity Analysis and Duality -> special/error cases -> output file and final executable.

6. Optional Bonus Work

Only attempt the bonus after the required project is complete and there is clearly enough time remaining. The brief offers up to 10 bonus marks for solving a non-linear problem such as $f(x)=x^2$ with an algorithm and explaining that code.

- Any member can volunteer for the bonus once their main responsibilities are under control.
- The bonus volunteer must also prepare and present/explain the bonus section in the final video.
- If Option B is used, the bonus volunteer can present only the bonus section after the main presenter finishes the required project. This avoids adding even more work to the sole presenter.
- Do not let bonus work delay integration, testing, the required video, or any of the 100-mark criteria.

7. Decisions to Make at the Team Meeting

- Choose Console App or Windows Forms App.
- Choose Video Option A or Option B.
- Assign the four roles. If Option B is chosen, make sure the sole presenter receives the lightest coding role.
- Create the GitHub repository and feature branches.
- Agree on shared model/tableau/solution classes before development starts.
- Choose who will edit/combine the final video.
- Only near the end, decide whether someone has enough time to volunteer for the bonus section.